Is It Deep Enough?
An analysis into Sac Fly Distance

Team 102

Abstract

Determining how far a ball has to travel for a sacrifice fly to successfully score the runner from third base has been a question for many baseball fans. In this paper, I create a sac fly line that gives fans a perspective on how far a ball has to travel for a successful sac fly. A logistic regression model is created to determine the probability of a successful sac fly based on the fielder's throwing ability, the distance the ball travels and the *pace per ninety feet* of the runner on third. The fielder's throwing ability is quantified by a novel metric I developed called *sniper*, which provides a value that combines a fielder's throwing strength alongside their accuracy in throws to home plate. With this model, it is now possible to compute the distance the ball must travel to have a specific probability of the corresponding sac fly being successful. A Shiny app is used to demonstrate how this can be displayed on the jumbotron during a baseball game to provide fans valuable information regarding the success of a potential sac fly.

1. Introduction

It is game 163 in the 2007 season with the Colorado Rockies playing the San Diego Padres. The winner of the game makes the postseason while the loser's season is over. It is a tied game at the bottom of the 13th inning where Colorado has men on third and first base with zero outs for Jamie Carroll, who hits a fly ball to right field (Link to the play). At this moment, there is one question that most baseball fans are asking: should the runner at third run home? It is difficult for fans to determine how far is far enough for a successful sac fly. Therefore, this project is focused on creating a model that displays on a jumbotron to display how far the ball needs to travel for a sac fly to likely be successful. In doing this, I created a new metric, *sniper*, to quantify the throwing ability of an outfielder, based on power and accuracy.

The inspiration behind this idea is from the National Football League (NFL). During NFL games, broadcasters superimpose a field goal line onto the field to alert the viewers how far the offense needs to advance before a field goal can realistically be successful. An example of this is visualized in Figure 1.

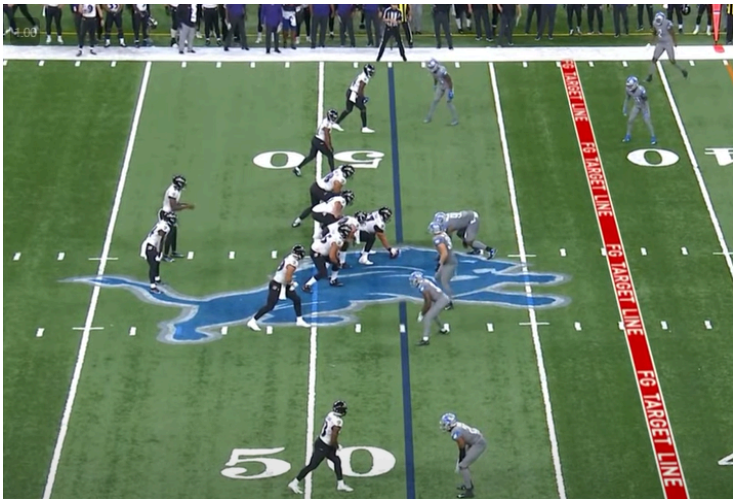


Figure 1: An image of an NFL broadcast showing the Baltimore Ravens lining up for a play against the Detroit Lions. They are aiming to pass the red Field Goal target line (Source)

The determination of the NFL field goal target line is dependent on the kicker's career high and is based on what would be considered reasonable for that kicker. This methodology will serve as the basis of my new model.

In terms of past research into fielders throwing abilities, Baseball Savant has *Fielding Run Value* [1]. This statistic focuses more on fielding abilities than throwing power and accuracy. There is also arm strength, but this metric only focuses on power, ignoring accuracy [2]. Given these limitations, a new metric would need to be created if I want to use arm strength and accuracy together.

2. Analysis

2.1 Data

The data from SMT includes player positions (in x, y coordinates), ball position (in x, y and z coordinates), play and lineup data for a season of minor league baseball. One key limitation with the data is that outs were not provided, so base-outs states must be inferred. Nevertheless, it still provides something to work with for a proof of concept. I also removed data past the eighth inning to ignore walkoff situations.

For the calculation of *sniper*, all throws to home in the data were used to maximize the amount of data and provide a better metric. For the Carry Distance model, I only used plays where a sacrifice fly could occur to ensure that the data used in the model is accurate to a sac fly situation.

2.2 Sniper Model Calculation

I use the basis of how *arm strength* is calculated to create a new metric called *sniper*. I chose to create a new metric instead of having both accuracy and velocity in the carry distance model because these factors will be correlated as both are related to the outfielder. Having a metric that combines both fielder strength and accuracy has uses beyond the carry distance model (see Appendix 2 for more details).

Arm strength is measured by taking the average of the top 10% of every player's throws [2]. I apply this same process to *sniper* calculations, focusing on the top 10% of hardest throws. I also look at the mean accuracy of those throws, which is the distance between the left corner of home plate and where the ball is caught. Choosing the left corner of home plate is sensible because this is where the catcher would want to catch the ball to ensure that they can easily tag the runner. I then centered and normalized these means and used Principal Component Analysis (PCA) to get the coefficients for this new metric. The range of this metric found inside of the data is (-2.753, 2.266).

Figure 2 depicts two plays, one showing a high positive *sniper* (2.266) and one showing a high negative *sniper* (-2.217). Figure 2(A) has the highest *sniper* metric of the outfield throws in the dataset. In this play, the fielder throws the ball and the catcher catches the ball at the left corner of the plate. In comparison, Figure 2(B) was one of the worst *snipers* in the dataset. Despite the similar velocity between the two throws, this throw causes the catcher to move further off of the plate, leading to a worse tag attempt towards home. Appendix 2 explains the creation of the *sniper* metric in more detail.

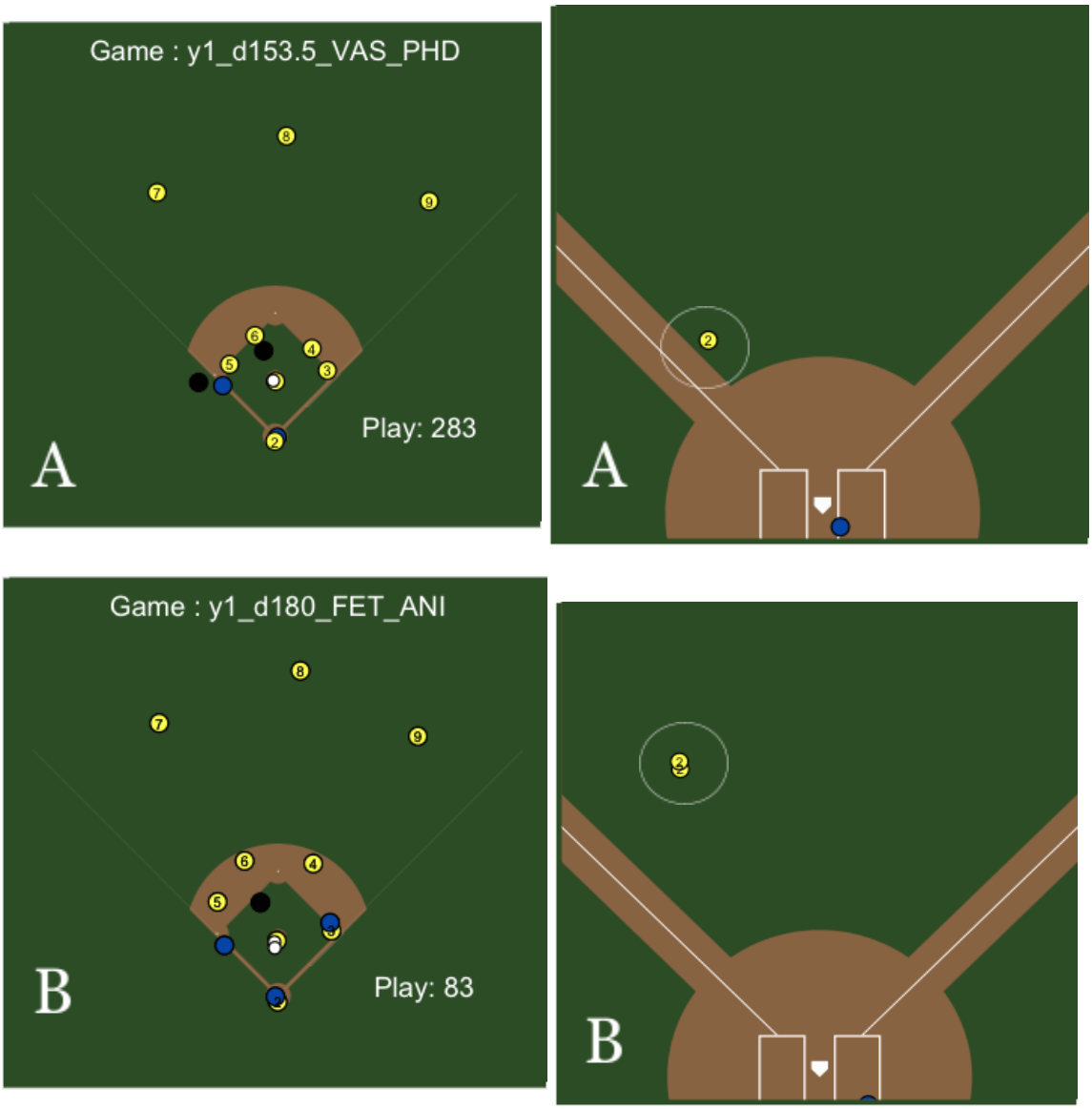


Figure 2: Examples of a good *sniper* (A) and a poor *sniper* (B). The blue dot is the runner and the yellow dot with number 2 is the catcher when they catch the ball.

2.3.1 Inputs for Carry Distance Model

Due to anonymization, I was unable to definitively determine whether or not a sac fly was successful or not. To overcome this, I judged whether the play was a close play by determining whether the catcher caught the ball before the runner crossed home plate. For the base runner, I take the mean of the top 10% of each player's *pace per ninety feet*. At that point, I assume the runner will be running at full speed.

Here are the inputs I am using to affect the model in terms of probability of a sac fly being successful:

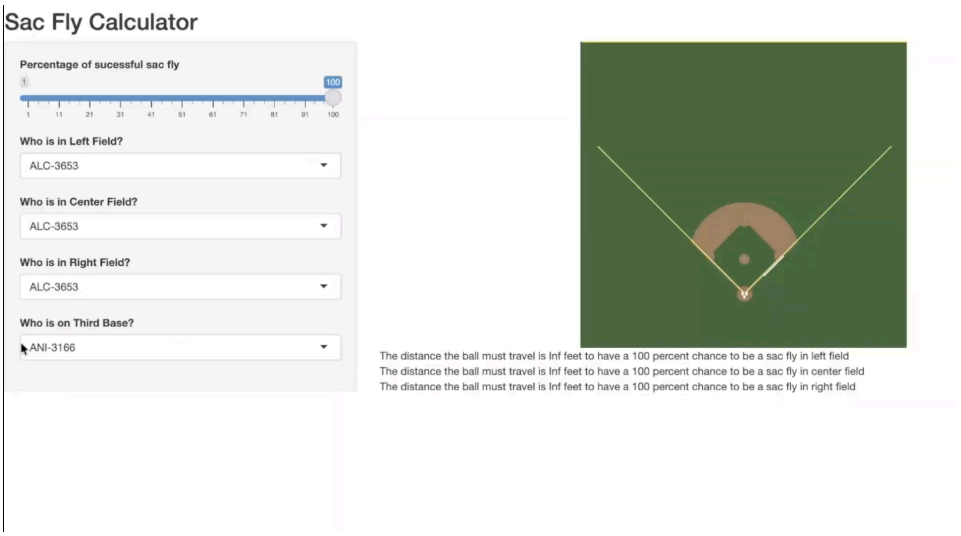
- The fielder's *sniper* metric
- How far the ball travelled.
- The pace per ninety feet of the runner on third

2.3.2 Model for Carry Distance

For any play, I have three different estimations of the probability of a successful sac fly, one for each outfielder. I use a logistic regression model to estimate the probability of a play being successful. This model provides coefficients (weights) for all three of the factors (*sniper*, *pace per ninety feet* and distance thrown) which is used to get the sac fly lines. A more in-depth explanation of the model is provided in Appendix 3.

3. Presentation

3.1 Shiny App



Video 1: Video example of the Shiny app

I created a Shiny app that allows you to add what players are in what part of the outfield and who is on third to help create the visual used on the jumbotron. Once these values are provided, the app uses the parameters found from the logistic regression model to add lines showing how

far the ball has to travel for there to be a *p%* chance for a sac fly to be successful (where p is also user-specified in the Shiny app). A hard cap of the maximum distance was added based on the deepest fields in MLB stadiums. The limits are 435 ft in center field (Daikin park), 350 ft in right field (Wrigley Field) and 355 ft in left field (Wrigley Field) [3]. One problem with the app is because of the lack of data, the model does give negative distance values when the set probability is low (less than 20%), but I believe works well when dealing with probabilities of at least 50% which is what we would really want to focus on. Video 1 is an example of the app in progress and here is the link to the Shiny app for usage (Shiny app).

3.2 Jumbotron Presentation

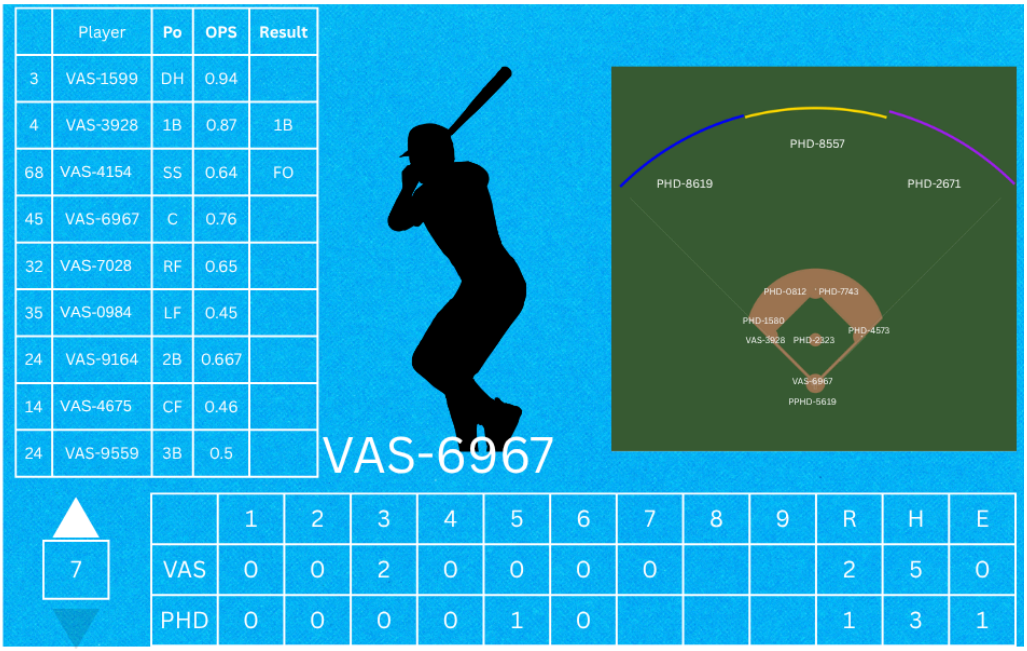


Figure 3: Example of jumbotron display using the play from Figure 2. Note: the OPS statistics were entirely made up in this case.

Figure 3 presents an example of how this model could be presented on a jumbotron. This image would appear whenever there is a sac fly situation. The lines are created using the Shiny app to provide a distance where there would be a the probability of a successful sac fly would be 80%. A more in depth analysis of how this line is found in Appendix 4. As seen in Section 3.1, this percentage can be changed if desired.

4. Conclusion, Next Steps and Further Applications

Overall, sac fly carry distance allows fans to easily see how far the ball has to go for a sac fly to reasonably occur. Potential next steps for this project include adding additional inputs that were not available in the SMT dataset to see if they have a significant effect on the model. Some examples of these potential inputs include field conditions, the current score of the game and current inning. In terms of model calculations with more data, I would use data from previous

seasons to train both the *sniper* and carry distance models and use those models for current season data.

4.1 Future Developments/ Model Limitations

One future development of this project is not having the three fielders have equal field presence. Another future development could be walkoff situations such as the one mentioned at the start of this paper. It would also be interesting to see whether sprint speed could be used as an alternative to *pace per ninety feet* as in theory, they both represent the same thing, that being how long the fielder has to get the ball to the catcher. One metric I was initially interested in adding was how far the runner has to run to catch the ball. While it would be interesting to see whether or not that has an effect on sac fly situations,this model will be calculated before the at bat. Another limitation is the current data only has 65 different players who have *sniper* metrics and 33 players who have running data.

4.2 Carry Distance Model for Player Development and Evaluation

Extending the methodology beyond the jumbotron, this carry distance model can also be used to aid in player development and evaluation. An example of this is showing players and coaches who should be throwing the ball home. This can also allow teams to get a better understanding of who may be a player that would be subbed off for a better defensive player.

This model also has a lot of benefits in terms of in-season scouting. If they know a specific fielder on the opposing team has a lower *sniper* metric than the other outfielders, then there may be some discussions about where to aim for in sac fly situations and having numbers to back up the claims. For example, if a manager knows that the opposing right fielder has a lower *sniper* than the left fielder, then the manager and hitting coach may tell their team to aim for right field in a sac fly situation.

4.3 Carry Distance Model for Broadcast

As mentioned at the start of the paper, the idea of the sac fly lines came from the field goal target lines from NFL broadcasts. Based on that, these sac fly lines are usable not just on jumbotrons, but also on national broadcasts. There are two ways I could see this being used in practice. One would be when the camera pans to the defensive formation before the at bat, overlaying the sac fly distance.

The second way I see broadcasters using it is when the camera pans to an outfielder during a fly ball, the broadcast would show the sac fly line or provide the probability of a successful sac fly occurring. Both of these methods are especially beneficial for more casual fans who may not know the specific details about baseball and allows for easier accessibility and growth of the game.

Acknowledgements:

I would like to thank Billy Fryer, Dr. Meredith Willis, Cameron Adams and David Awosoga for helping with the design of the project, paper editing and providing tutorial code which helped with getting ball travel distance. I would also like to thank my parents and Dr. Nathaniel Stevens for proofreading the paper and providing edits.

Works Cited:

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Appendix 1: Explanation of Out Detection for the Carry Distance Model

I started by grouping together plays with the same batter and runners on base. The reason I grouped the plays together was due to problems with the data set where plays were not ordered correctly.

I know that there are 0 outs at the start of every game, and every half inning. I also know that the last out occurs on the last play of the half inning, so the main thing is to find the other two outs. Then from there, I looked at the last play to see whether or not there was a home run. If there was a home run on that play, then I knew there were no outs at all.

Next, I check to see if the batter is on base in the next play. If they are not, then I know that the batter is out. Finally, if the batter got out and a runner on first or second base disappeared, then I assume they are out as well, as it is extremely unlikely that someone scores from first or second on a sac play.

Appendix 2: Sniper Metric Details and Use Cases

As mentioned earlier, I decided that for each player to only select plays where the velocity of the throw at its maximum (so right after it was thrown) was in the 90th percentile of the fielder's throws to home plate. Once this was done, I took the mean of the velocity and the distance from the left corner of the plate where the catcher caught the ball and used PCA to get the coefficients. I used PCA to combine the two metrics into one single value (*sniper*). The weighting was decided by the first principal component. This model was successful in providing values as it explained approximately 55% of the cumulative variance of the two variables. The equation was the following: $s = 0.707 * v - 0.707 * a$, where v is the velocity and a is the distance off the plate it was caught. I was surprised that both estimates were the same, and I would be interested to see if this is still the case if I had more data. In the future, *sniper* could have great use both in the analytics sector of Baseball, but also in player development and evaluation. Coaches could show a player how their *sniper* change over the span of their career. This can also be used in terms of drafting (if these models are reworked for the high school or NCAA levels) or determining which free agent to sign. More information allows for better judgement calls to be made.

Appendix 3: Technical Details for the Carry Distance Model

The equation provided by the logistic regression model is the following:

$$logit(p) = \alpha + \beta s + \gamma r + \theta d$$

Where $logit(p) = ln(p/(1-p))$, p is the probability of a sac fly being successful, s being the fielders *sniper* metric, d being the distance the ball went and r being the runners pace per ninety feet. The Greek letters are the parameter estimates that are found by fitting the model to the data.

Term	**Estimate**	**Std.error**	**Test Statistic**	**p.value**
(Intercept)	-1.972	3.713	-0.531	0.595
sniper	-0.072	0.333	-0.217	0.829
distance_thrown	0.009	0.015	0.603	0.547
pace_per_ninety	0.007	0.031	0.234	0.815

Table 1: Summary output of the carry distance model.

Note that the p-value for each estimate is greater than 0.05, which means that the estimates are not significant. This means that the estimates for these parameters have a lot of variance and the true value of the underlying parameter could be 0, which would very problematic in determining d. I believe most of this is caused by the small sample size as it does not make sense for how far the ball travels to not be a significant factor in determining whether a sac fly is successful. If I had more data, then I believe all of these factors would be significant, leading to less variance within the estimates.

Appendix 4: Shiny app technical details

For this Shiny app, I use the coefficients found in the logistic regression model and instead of trying to find the p given s, r, and d, I solve for d given s, r, and p. Therefore, the equation from Appendix 3 is reworked into:

$$d = \frac{logit(p) - \alpha - \beta s - \gamma r}{\theta}$$

Then the next problem was to get the cartesian coordinates for the three lines (one for each fielder). The only thing we know is that $d = \sqrt{x^2 + y^2}$ where x and y are the cartesian coordinates on the plot. One solution I came up with was having constituent coordinates for x. However, this did not work due to the range of x between the foul lines not being consistent over every of y. Notice how the outfield gets wider the deeper it is. To remedy this, I used Polar coordinates as all I need are the angles of the foul lines, which are $\frac{3\pi}{4}$ for the left foul line and $\frac{\pi}{4}$ for the right foul line. Then from there I can split that range into three equal sections and use the angles τ in each section to cartesian coordinates, using the equations $x = dcos(\tau)$ and $y = dsin(\tau)$. For left field, τ ranges from $\frac{3\pi}{4}$ to $\frac{7\pi}{12}$, center field ranged from $\frac{7\pi}{12}$ to $\frac{5\pi}{12}$ and right field ranged from $\frac{5\pi}{12}$ to $\frac{\pi}{4}$. This way, even as the distance changes, all three lines should be of similar length and do not pass the foul lines.